- **SYSTEM DESIGN:-**

  - System design is the process of defining the architecture, components, modules, interfaces and data for a system to satisfy specified requirements. System design is therefore the process of defining and developing systems to satisfy specified requirements of the user.
  - It is process of planning new system or replacing an existing system.
  - It is done by defining its component and module to satisfy the requirements
  - It is focus on how to accomplish the objective of the system.
  - System design is highly creative phase which focuses on how to make a system that is fully functional, reliable, easy to understand, verifiable, consistent and easy to operate.

System design objectives

To produce a good system design, the following are the design objectives:

1. **Changeability**
   Changeability is the ease with which a system may be changed. The system should be modifiable depending on the changing needs of the users or changes in environment or policies etc.
2. **Correctness**
   The system design should be correct. Correctness is the extent to which a system design is free from defects i.e. fault-free. The extent to which system design meets its specified requirements & user expectations also comes under correctness.
3. **Efficiency**
   Efficiency is the extent to which a system performs its intended functions with a minimum consumption of computing resources. The best use should be made of people and equipment. Efficiency also involves the accuracy, timeliness and comprehensiveness of the system's output.
4. **Reliability**
   The system design should be reliable. Reliability is the ability of a system to perform a required function under stated condition for a stated period of time. The system should display proper error messages.
5. **Security**
   Security is an important aspect of the design. Security implies/includes the following factors:

● **Confidentiality**

Some important information needed for the firm, has to be kept out of the reach of competitors. The system has to ensure that only authorized staff have such information.

● **Privacy** The system must guard against any unauthorized access to the information about the individual employee.

● **Data Security** Measures are needed to guard against alternation or destruction of data.

6. Expandability

The system should be able to cope with seasonal or cyclical variation/fluctuation in volume/load.

The designed system should also be:

● User-friendly

● Program-friendly

● Ensure that system features meet user requirements

● Cost-effective

● Easy to use and

Logical and physical design

- Logical and physical design are two distinct phases in the process of creating a system, typically used in software or systems.
- They represent different levels of abstraction and detail in the design process.

- ### **Logical Design**:

    - **Abstraction Level**: Logical design deals with the abstract representation of the system, focusing on what the system does rather than how it is implemented.
    - **Conceptualization**: In this phase, the emphasis is on understanding and defining the requirements and functionalities of the system. It involves creating a conceptual model that captures the essential features and relationships within the system
    - **No Platform Dependencies**: Logical design is platform-independent. It does not concern itself with specific hardware or software components but instead focuses on defining the system's functionality, structure, and behavior..

- **Techniques**: Techniques such as data flow diagrams, entity-relationship diagrams, and use case diagrams are commonly used in logical design to represent the system's requirements and functionalities.

- ## **Physical Design**:
- P D describe how the proposed system deliver the capabilities specified in logical design
  - **Abstraction Level**: Physical design deals with the concrete implementation of the system, focusing on how the system will be realized using specific hardware and software components.
  - **Implementation Planning**: In this phase, the emphasis is on selecting the appropriate technologies, platforms, and architectures to meet the requirements defined in the logical design phase.
  - **Platform Dependencies**: Physical design is platform-dependent. It involves specifying the hardware infrastructure, software components, networking configurations, and other technical details required to implement the system.

Structured design methodology

- The aim of design methodology is to provide guidelines to the designer during the design process.
- Structured design was proposed by Stevens, Myers and Constantine to provide methods for dealing with the complexity as programs grow larger.

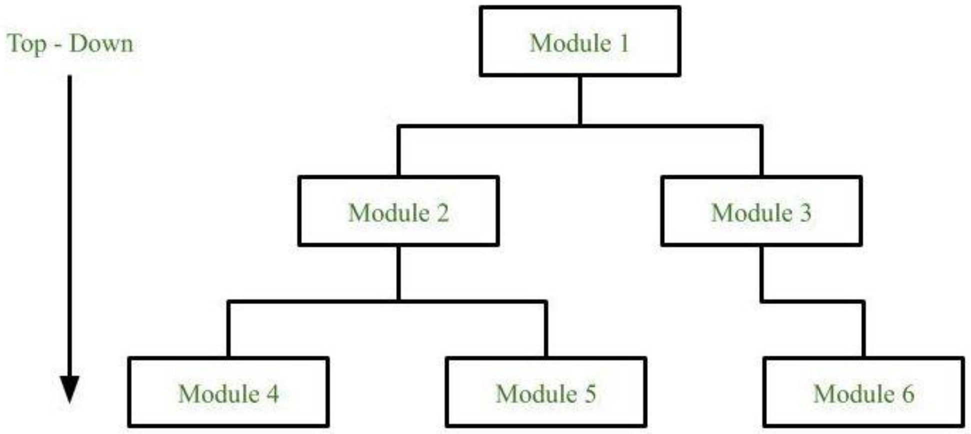
In this structured design methodology the system is designed in the following steps:

- Identification of input and output
- Description of functional aspects of the system
- Representation of system in DFDs
- Formulation of data dictionary
- Documentation of structured design by structured chart

Software Design Approaches

Top-down approach:

- Each system is divided into several subsystems and components.
- Each of the subsystems is further divided into a set of subsystems and components.
- This process of division facilitates forming a system hierarchy structure.

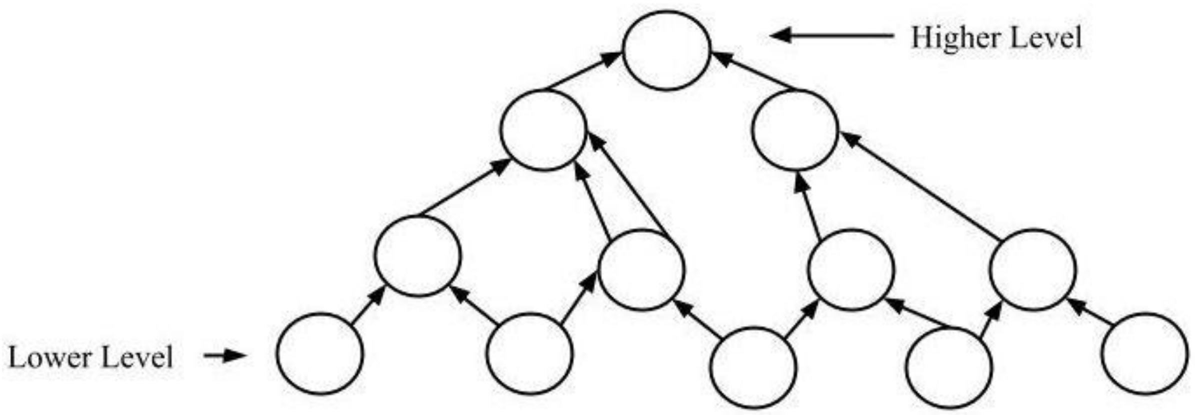


Advantages of Top-down approach:

- The main advantage of the top-down approach is that its strong focus on requirements helps to make a design responsive according to its requirements.
- Any change can be done easily
- Reusability

Bottom-up approach:

- The design starts with the lowest level components and subsystems. By using these components, the next immediate higher-level components and subsystems are created or composed.
- The process is continued till all the components and subsystems are composed into a single component, which is considered as the complete system.



Advantages of Bottom-up approach:

- Increase the collaboration among the team member
- Fault localization is easy
- Easy to Maintains

MODULARISATION:-

- Modularisation in the decomposition of a system in to manageable components called module where each module can be coded separately.
- **A module** is a unit of program or function which associated attribute element data performing a specific task. In C language modules are called function, in Java and C++ modules are called classes.

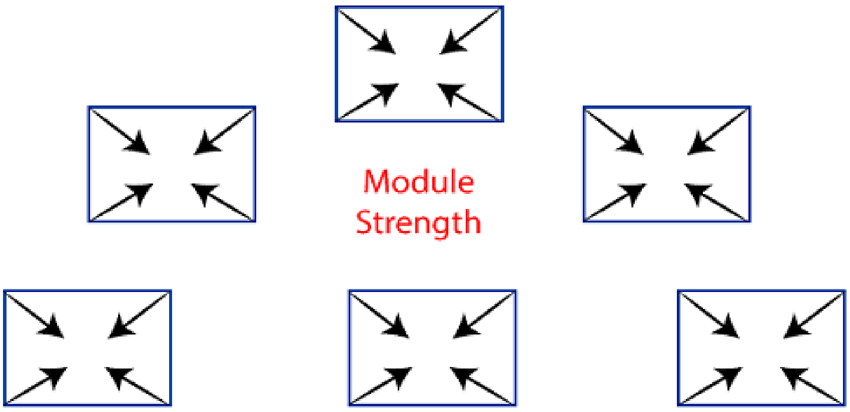
In terms of cohesion and coupling, the main goal of structured design is to divide a problem into modules such that the coupling between different modules is reduced while the cohesion of different modules is enhanced.

It is necessary to understand the concept of coupling and cohesion for the following purposes/reasons:

● To understand the way the modules interact with each other.
● To design to improve changeability.
● To minimize the number and complexity of interconnections between the modules.

COHESION:-

- cohesion defines to the degree to which the elements of a module belong together. Thus, cohesion measures the strength of relationships between pieces of functionality within a given module. For example, in highly cohesive systems, functionality is strongly related..



Cohesion= Strength of relations within Modules

Types of cohesion are:

1. Coincidental cohesion Worst (lowest cohesion)
2. Logical cohesion
3. Temporal cohesion
4. Procedural cohesion
5. Communicational cohesion Best (highest)

1. Coincidental Cohesion.

- ● This weak degree of cohesion is found in a module whole elements have no meaningful relationship to one another.
- Coincidental cohesion can occur when an existing program is modularized by breaking into pieces, where pieces themselves are different modules.

2. Logical Cohesion.

- It is the next higher level of cohesion where several logically related functions or data elements are placed in the same module.
- e.g., Read all kind of input from tape, disk, communication port.
- Print appropriate messages for all.

3. Temporal Cohesion.

- It is same as logical cohesion except that the elements are executed at the same time.
- All actions related to initialization or termination of a processcan be kept in one module.
- e.g., print final totals and close all files.

4. Procedural Cohesion. -

- A module is said to possess procedural cohesion if the set of all function of the module are all sort of procedure in which a certain order for achieving an objective.
- Here the instructions in a module are related to each other through flow of control.
- e.g., The algorithm for decoding a message.

5. **Communicational Cohesion**.

- When modules operate on the same input data or contribute towards the same output data.
- Communication cohesion module may be performing more than one function

<u>Coupling:-</u>

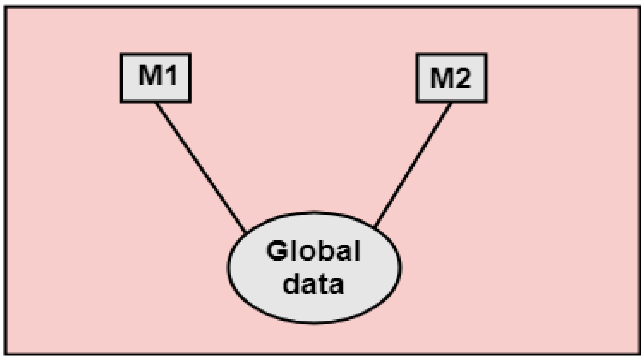
- ✓ **Coupling** is a measurement that defines the level of inter-dependability between modules of a program.
- ✓ In other words, it is a kind of measurement that defines how software components depend on each other and interact.
- ✓ Lower coupling indicates a better software design.

TYPES OF COUPLING

1. Content Coupling
2. Common coupling
3. Control coupling
4. Data coupling Best

- **Content Coupling** is the highest level of coupling where one module relies heavily on the internal details of another module. This occurs when one module has direct access to the internal variables or data of another module. In other words, if there is any change in one module, it will necessitate a change in the dependent module as well.

- **Common Coupling,** also known as Global Coupling, occurs when multiple modules share common global data. Changes to this global data will affect all modules that use it.

- 

- **In control coupling,** one module influences or directs the flow of execution in another module by passing control information.
- **Data coupling** occurs between modules when they communicate by passing only necessary data between them.

Tool for Structured Design

Structured chart is the fundamental tool used for the structured design.

Structured chart is a graphical representation which shows:

- System partition into modules.
- Hierarchy of component modules.
- Relation between processing modules.
- Interaction between modules.
- Information passed between modules.

Structured charts, use standardized symbols to represent different elements of the program's structure and flow of control. Here are some commonly used symbols in structured charts:

Start/End Symbol:

- Represents the beginning or end of the program. Usually depicted as a rounded rectangle or oval shape.

Process/Action Symbol:

- Represents a specific action or operation performed within the program. Usually depicted as a rectangle.

Input/Output Symbol:

- Represents input or output operations, such as reading data from a user or displaying information. Usually depicted as a parallelogram.

<u>Loop Symbol</u>:

- Represents a loop or repetition structure in the program. Usually depicted as a curved or circular arrow.

Symbol	Name	Function
	Start/end	An oval represents a start or end point
→	Arrows	A line is a connector that shows relationships between the representative shapes
	Input/Output	A parallelogram represents input or output
	Process	A rectagle represents a process
	Decision	A diamond indicates a decision

- ## <u>Form-Driven methodology</u>

- The Form-Driven methodology is a software development approach that focuses on designing and implementing forms as the primary means of interaction between users and the system.

- Below are the steps typically involved in the Form-Driven methodology, along with a process:

  - **Requirements Gathering:** Gather requirements from stakeholders and users regarding the functionality and features expected in the software system.

- **Form Design & Mockups:** Design user interface forms based on the gathered requirements. Create mockups or prototypes to visualize the forms.
- **Iteration & Feedback:** Present the form designs to stakeholders and users for feedback. Iterate on the designs based on received feedback to refine them.
- **Functional Specification:** Develop detailed functional specifications based on the finalized form designs. These specifications outline the functionality and data inputs/outputs for each form.
- **Development:** Implement the forms and underlying logic required for user interaction based on the functional specifications.
- **Testing & Validation**: Test the forms rigorously to ensure they function correctly and meet the specified requirements. Validate the forms through usability testing and functionality testing.
- **Refinement & Enhancement:** Refine and enhance the forms based on feedback and testing results. Iterate on the forms to improve user experience and functionality.
- **Integration & Deployment**: Integrate the forms into the larger software system and deploy the system for end-users to access.
- **Maintenance & Updates**: Provide ongoing maintenance and updates to the forms as needed. Monitor user feedback and usage data to continuously improve the forms over time.

Forms Design

- Form is that which serves as a tool for taking data from the people and giving information of the system.
- Form is used for gaining the information and taking decision.
- Both forms and reports are the product of input and output design and are business document consisting of specified data. The main difference is that forms provide fields for data input but reports are purely used for reading.
- Forms design is a crucial aspect of user interface design, particularly in applications and systems where users need to input information.
- Forms are used to collect data from users, and a well-designed form ensures that this process is efficient, intuitive, and error-free.
-

For example, order forms, employment and credit application, etc.

- During form designing, the designers should know –
- the purpose of the form or report
- who will use them
- where would they be delivered
- During form design, automated design tools enhance the developer's ability to prototype
- Forms and reports and present them to end users for evaluation.

Objectives of Good Form Design

- A good form design is necessary to ensure the following −
- To keep the screen simple by giving proper sequence, information, and clear captions.
- To meet the intended purpose by using appropriate forms.
- To ensure the completion of form with accuracy.
- To keep the forms attractive by using icons, inverse video, or blinking cursors etc.
- To facilitate navigation.

Classification of forms:-

There are three primary classifications:-

1. **Action Form:-** An action form requests the user to do something and get action.

- **E.g. application form, purchase order, shop order.**

Characteristics of action form:-

o These type of forms are used to order, instruct and organize.

o They result in getting something done.

o These forms go from one place/person to another.

o Aim at achieving quick result.

o The forms are filled by the users and sent for processing.

Significance of action form:-

o Clearly state the purpose.

o They reduce unnecessary detailing of data.

o They facilitate quick reading of the lines across the form.

o Data is presented in a logical sequence with proper space for the required information. E.g. signature etc.

2. <u>Memory Form:-</u>A memory form is a record of historical data that remains in a file, is used for reference and serves as control on keys details.

E.g. bond registers, inventory record.

Characteristics of memory form:-

- o Keeps the record
- o Represents historical data
- o It is used for future reference
- o Kept at one place

Significance of memory form:-

- o The columns and rows or specified labelling's help in clear-cut presentation.
- o Less time consuming
- o Data entry become easier
- o No confusion when information is placed under proper headings.

3**. <u>Report Form:-</u>**A report form guides supervisory another administrators in their activities. It provides data on a project or a job.

E.g. balance sheet, profit and loss statements etc.

Characteristics of report form

- o Provide descriptive information
- o Present summary of a project
- o Can be used as a basis for decision-making
- o Used for managerial purpose

Significance of report form

- o The title of the form clearly states the purpose of the form
- o Serves as a basis for decision- making
- o Provide summary and the managerial information

Requirements/ factors of forms Design:-

1. **Understandable heading:-** The title of the form should clearly indicate the purpose of the form. It should be as simple as possible. The column and rows should be specifically labelled.

2. **Easy usage:-** The form must be easy to use and fill. It should not look complicated to complete.

3. **Physical attributes:-** The forms color, margin, layout etc. are also an important part of form design. Stripping and shading can be used for highlighting parts of a form.

4. **Size and arrangement:-** Size and shape of the form should be convenient for handling , filling, sorting etc. A prominent location and shape should be marked for important items and signature etc.

5. **Logical flow of information:-** The entries on a form should ensure a logical sequence of the related material.

6. **Easy data entry:-** The form must be designed in such a way so that the data entry becomes easier.

7. **Sufficient instructions for the user:-** Sufficient instructions should be included in the form to inform the user how to use the form.

8. **Efficient:-** The form must be cost effective and unnecessary data must be eliminated.

Types of Forms

1. Flat Forms

- It is a single copy form prepared manually or by a machine and printed on a paper. For additional copies of the original, carbon papers are inserted between copies.
- It is a simplest and inexpensive form to design, print, and reproduce, which uses less volume.

2. Unit Set/Snap out Forms

- These are papers with one-time carbons interleaved into unit sets for either handwritten or machine use.
- Carbons may be either blue or black, standard grade medium intensity. Generally, blue carbons are best for handwritten forms while black carbons are best for machine use.

3. Continuous strip/Fanfold Forms

- These are multiple unit forms joined in a continuous strip with perforations between each pair of forms.
- It is a less expensive method for large volume use.

4. No Carbon Required (NCR) Paper

- They use carbonless papers which have two chemical coatings (capsules), one on the face and the other on the back of a sheet of paper.
- When pressure is applied, the two capsules interact and create an image.

Layout considerations:-

When a form is designed, it considers all the items to be included on the form and the maximum space to be reserved. A list is prepared and checked to make sure that it has the required details.

Various layout considerations in forms design are:-

1. Form title
2. Form number
3. Data classification and zoning
4. Lines and captions
5. Box design
6. Size and shape of entry spaces
7. Ballot box and check-off designs
8. Paper selection
9. Cost considerations
10. More guidelines

1. Form title:-Form title should be brief and descriptive that shows what the form is and what it does. Generally the title is placed in the centre at the top of the form. Long titles should be avoided.

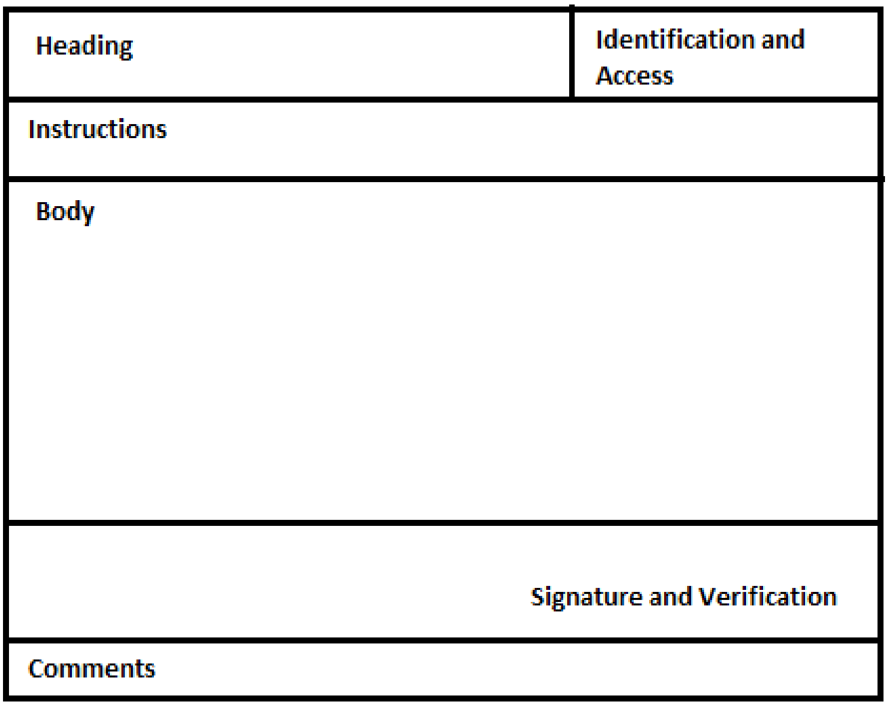
2. Form number:- A form number identifies the form and serves as a reference. It provides a control check to keep the record.

3. Data classification and zoning:- All the data items that must go on the form should be listed and classified into logical grouping. After the items are classified into a logical grouping the data groups are placed in appropriate areas/zones.

4. Lines and captions:-In designing forms, use of lines enables the human eye to read and write data.

- Caption serves same as a column heading. It specifies what information to write in the space provided.
- Lines guide and separate, whereas captions guide and instruct.

5. Box design:-Whenever possible, the form should be designed using the box with captions in the upper left corner



6. **<u>Size and shape of entry spaces</u>:-**There must be sufficient space to allow for data. The kind of data will determine the shape.

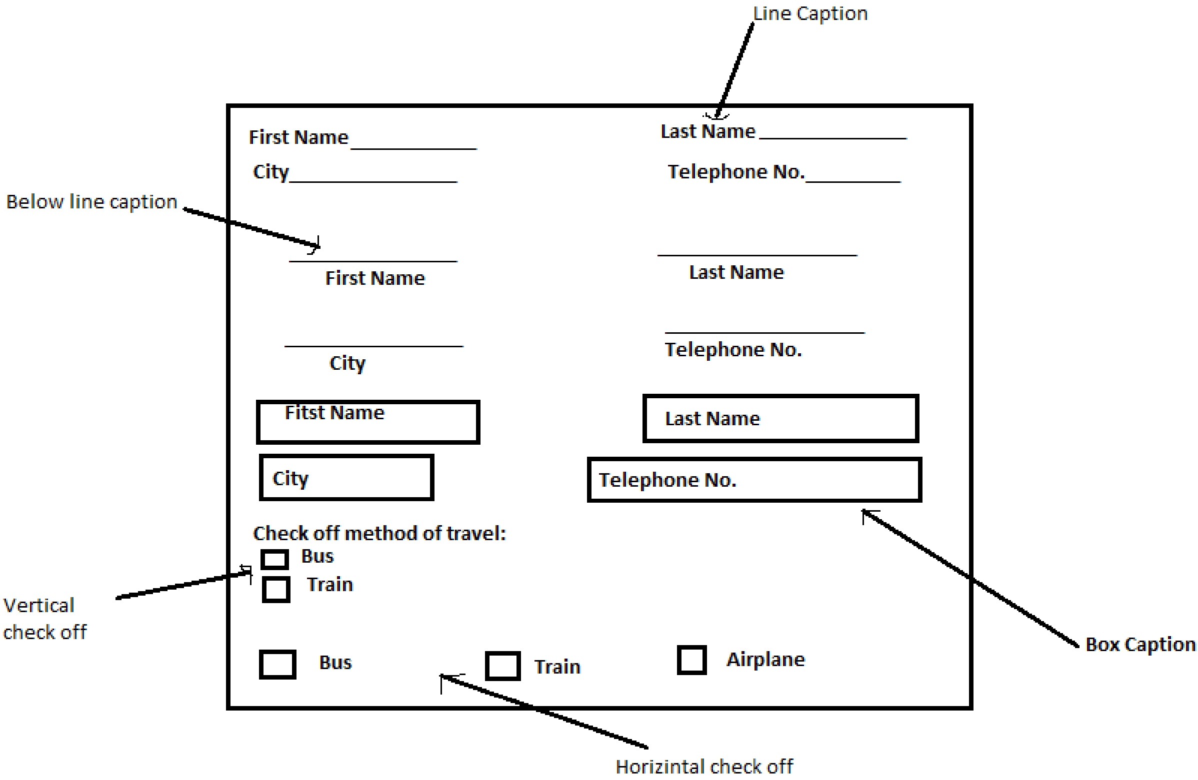
7. **<u>Ballot box and check-off designs</u>**:- For questions to be answered in yes or no, use of ballot or check-off boxes can greatly reduce the amount of writing.

8. <u>Paper Selection:-</u>Forms may be printed on paper of different colours, grades and weights. Colored paper or colored printing on white paper is used to distinguish among copies and facilities sorting copies.

9. **<u>Cost Considerations:-</u>**Various cost factors can be considered before the final decision to produce a form.

10. **<u>More guidelines:-</u>**

- Make forms easy to fill in.
- Ensure that forms meet the required purpose.
- Design forms to ensure accurate completion.
- Make forms attractive.



<u>FORMS CONTROL:-</u>

The first step in forms control is to determine whether a form is necessary. Managing the hundreds of forms in a typical organization requires a forms control program.

Form control is a procedure for:-

1. Providing improved and effective form

2. Reducing printing costs

3. Securing adequate stock at all time

The first step in a procedure for forms control is to collect, group, index, stock and control the forms of the organization. Each form is identified and classified by the function it performs and whether it is a flat form, a snapout form or something else. Once classified a form is evaluate by the data. It requires where it is used, how much it overlaps with other forms.

Input Design

In an information system, input is the raw data that is processed to produce output. During the input design, the developers must consider the input devices such as PC, MICR, OMR, etc.

Therefore, the quality of system input determines the quality of system output. Well-designed input forms and screens have following properties −

- It should serve specific purpose effectively such as storing, recording, and retrieving the information.
- It ensures proper completion with accuracy.
- It should be easy to fill and straightforward.
- It should focus on user's attention, consistency, and simplicity.
- All these objectives are obtained using the knowledge of basic design principles regarding −
  - What are the inputs needed for the system?
  - How end users respond to different elements of forms and screens.

Objectives for Input Design

The objectives of input design are −

- To design data entry and input procedures
- To reduce input volume
- To design source documents for data capture or devise other data capture methods
- To design input data records, data entry screens, user interface screens, etc.
- To use validation checks and develop effective input controls.

Data Input Methods

It is important to design appropriate data input methods to prevent errors while entering data. These methods depend on whether the data is entered by customers in forms manually and later entered by data entry operators, or data is directly entered by users on the PCs.

A system should prevent user from making mistakes by −

- Clear form design by leaving enough space for writing legibly.
- Clear instructions to fill form.
- Clear form design.
- Reducing key strokes.
- Immediate error feedback.

Some of the popular data input methods are −

- Batch input method (Offline data input method)
- Online data input method
- Computer readable forms
- Interactive data input

Output Design

The design of output is the most important task of any system. During output design, developers identify the type of outputs needed, and consider the necessary output controls and prototype report layouts.

Objectives of Output Design

- To develop output design that serves the intended purpose and eliminates the production of unwanted output.
- To develop the output design that meets the end users requirements.
- To deliver the appropriate quantity of output.
- To form the output in appropriate format and direct it to the right person.
- To make the output available on time for making good decisions.
- The following media devices are available for providing computer based object.
    1. MICR Readers
    2. Line, Matrix, Daisy Wheel Printers
    3. CRT Screen Display
    4. Graph Plotters
    5. Audio Response

Types of outputs −

External Outputs

Manufacturers create and design external outputs for printers. External outputs enable the system to leave the trigger actions on the part of their recipients or confirm actions to their recipients.

Some of the external outputs are designed as turnaround outputs, which are implemented as a form and re-enter the system as an input.

Internal outputs

Internal outputs are present inside the system, and used by end-users and managers. They support the management in decision making and reporting.